

Freiburg · 2026 September 17

Seeing every tree in the forest

Structure-aware and promptable tree segmentation
in 3D forest LiDAR point clouds

Trung Thanh Nguyen

Nagoya University · RIKEN, Japan



ABOUT ME

Trung Thanh Nguyen



*M.Sc. graduation
Nagoya University, Japan (2024)*

EDUCATION



2022 Bachelor

Hanoi Univ. of Science and
Technology, Vietnam



2024 Master

Nagoya University, Japan



2027 PhD (Expected)

Nagoya University

Nagoya University, Nagoya, Japan

PhD Candidate, Graduate School of Informatics, Department of Intelligent Systems

RIKEN, Kyoto, Japan

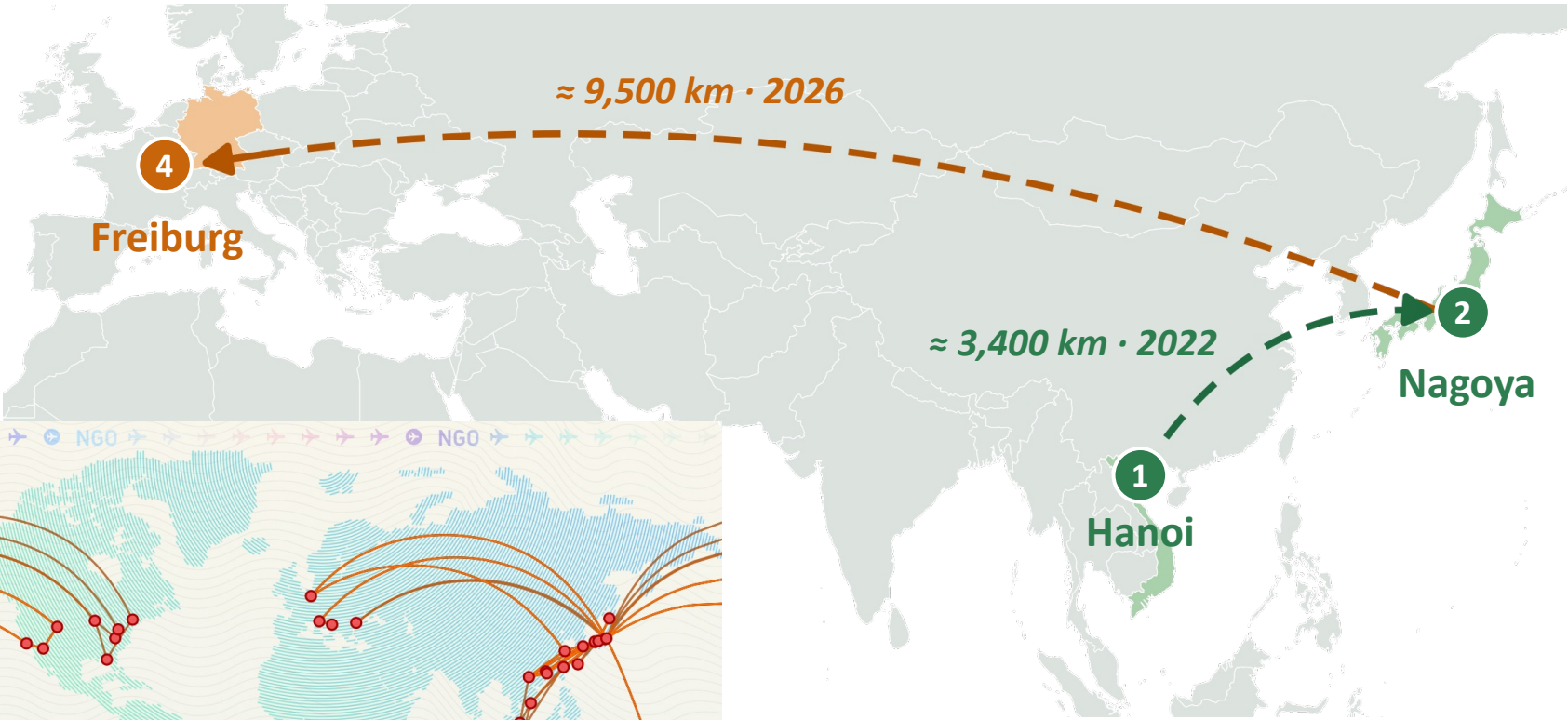
- Student Trainee
- Guardian Robot Project
- Multimodal Data Recognition Research Team

University of Freiburg, Germany

- Rising Star Fellowship
- Cluster of Excellence Future Forests
- Host PI: Prof. Teja Kattenborn

ABOUT ME

From Hanoi, to Nagoya, to Freiburg

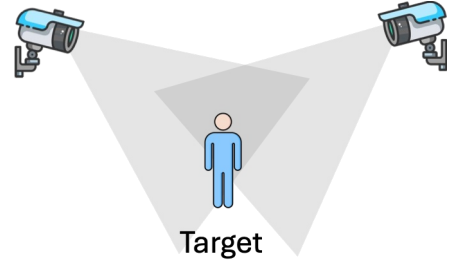


Nagoya University $\leftrightarrow$ RIKEN GRP
 $\approx 120 \text{ km}$



ABOUT ME

My research at a glance



PhD · NAGOYA & RIKEN

Human Actions from Multiple Sensors

#Keyword: Action Recognition, Multi-view, Multi-modal, Open-vocabulary



UNIVERSITY of FREIBURG

3D Forest Point Clouds 3dtrees.earth

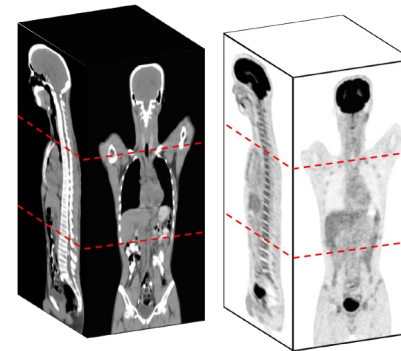
#Keyword: 3D Trees, 3D Point Clouds, Instance/Semantic Segmentation



UCLA, USA

4D Point Cloud Representations

#Keyword: 4D Point Cloud, Representation Learning



AI4LIFE · HUST, VIETNAM

Medical Imaging and VLMs

#Keyword: Diffusion Model, Report Generation, CT/PET Imaging

BACKGROUND

Forests are 3D, and ecology needs them tree by tree

31%

of Earth's land surface is covered by forest



Biomass and carbon

Stem and crown volume of each tree



Crown architecture

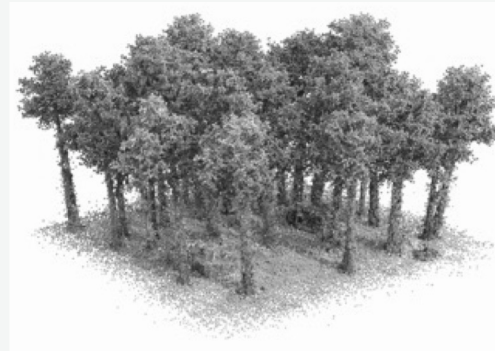
Canopy layers between neighbours



Health and biodiversity

Species, foliage vs. wood, forest condition

Joint Semantic + Instance Segmentation



Terrestrial,
Drone LiDAR Point Cloud

One label Per-tree,
+ Ground / Wood / Leaf

Every downstream trait starts from an accurate per-tree mask

BACKGROUND

Forest LiDAR is much harder than indoor 3D scenes



Massive Scale

Hundreds of millions of points per hectare at high acquisition density



Irregular Density

Occlusion and sensor geometry leave the upper canopy and far stems sparse



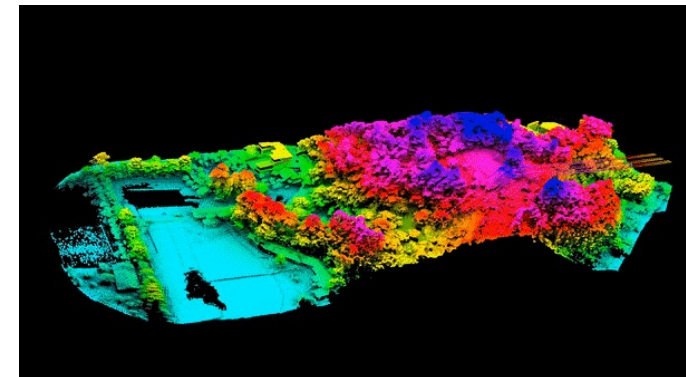
Entangled Canopies

Overlapping crowns make neighbouring trees hard to separate



Class Imbalance

Leaf points dominate;
Ground and Wood are much rarer



BACKGROUND

Existing methods and what is still missing

Clustering-based

ForAINet · TreeLearn · SegmentAnyTree

- Sparse-conv U-Net + mean-shift / DBSCAN grouping
- Hand-crafted grouping breaks in dense canopies

Query-based

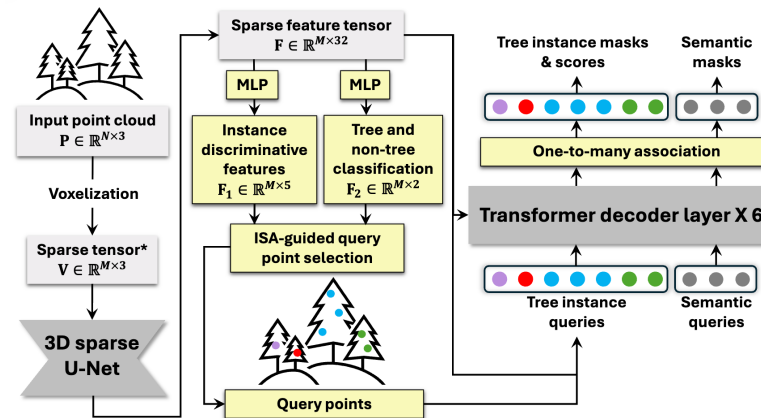
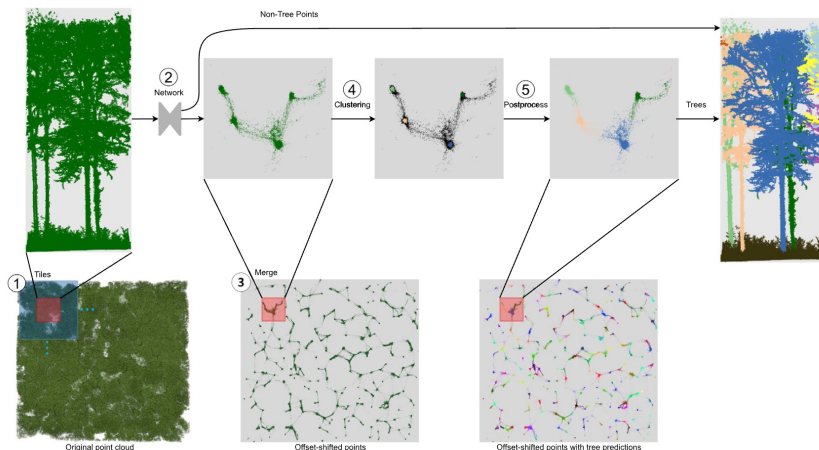
OneFormer3D · ForestFormer3D

- Transformer-based: Global attention + mask queries
- State-of-the-art accuracy

Three Gaps

- 1 Inference Cost**
Quadratic attention: slow and memory-hungry on large scenes
- 2 No Forest Priors**
Generic queries ignore canopy geometry, miss weak trees
- 3 No User Control**
All-or-nothing output: cannot pick or fix one tree

- Gaps 1–2 → ForestMamba
- Gap 3 → SelectAnyTree



* Represents non-empty voxels as a sparse feature tensor, where M is the number of voxels containing actual points.

OUTLINE

Two steps: faster automatic, then interactive

PART 1

ForestMamba

BMVC 2026

Can forest structure make automatic segmentation both more accurate and cheaper?

- 1 Vertical-priority sparse Mamba encoder
- 2 Treetop-seeded queries
- 3 Linear-time Mamba query decoder



PART 2

SelectAnyTree

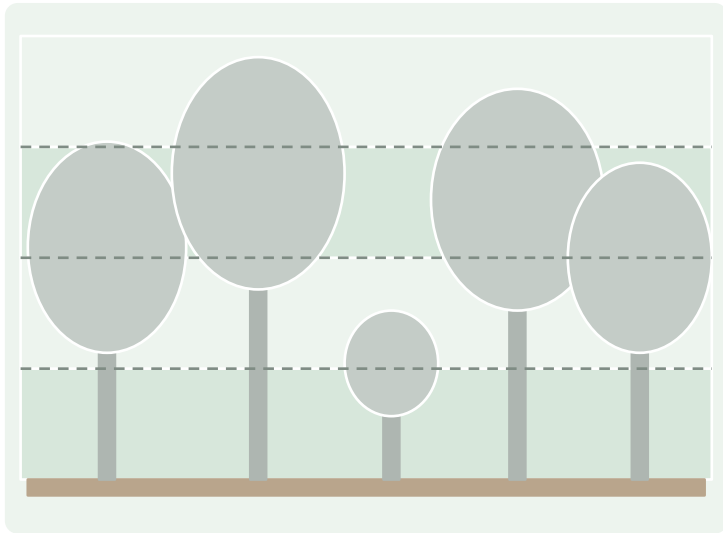
Under Review

Can a user select, and correct, any single tree with a few clicks?

- 1 Encode the scene once, reuse for every click
- 2 Click-to-query prompt encoder
- 3 Treetop-seeded as a free first click

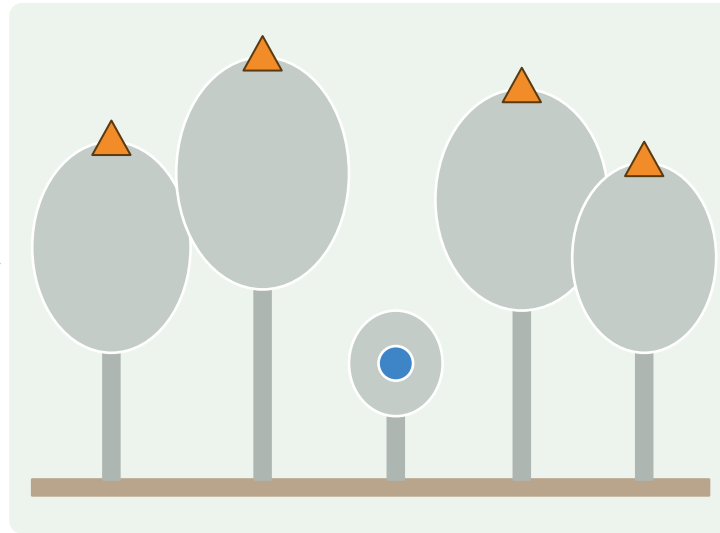
ForestMamba: the idea

In: LiDAR points (x, y, z) **Out:** every tree, plus ground / wood / leaf



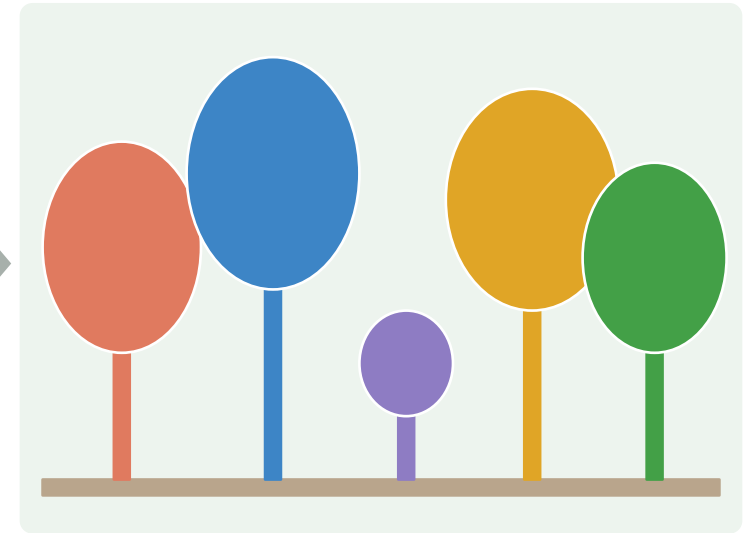
1 Read the Forest

Layer by layer, from the ground up



2 Mark the Treetops

One marker per tree, plus extra ones for small trees



3 Grow Each Tree

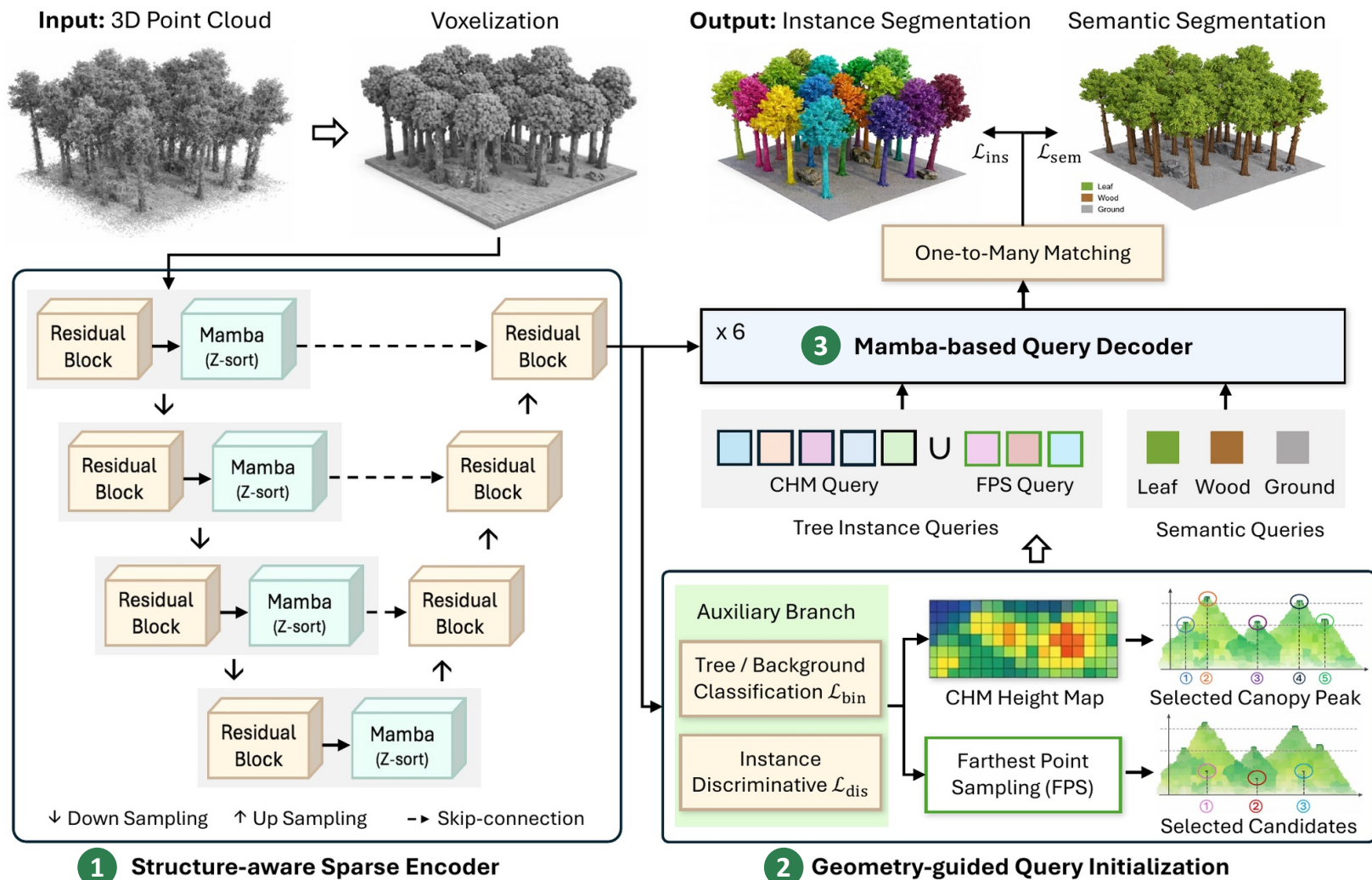
Each marker collects the points of its tree



In Plain Words

ForestMamba uses two things every forester knows: Trees grow upward, and every tree has a top

ForestMamba



1 Sparse Encoder

Sparse residual blocks + Mamba;
voxels ordered ground-to-canopy

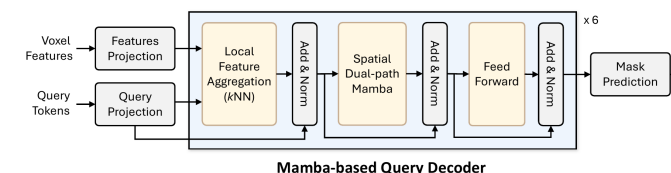
2 Geometry-guided Query Initialization

Auxiliary branch (tree / background)
→ CHM peaks + FPS queries

3 Mamba-based Query Decoder

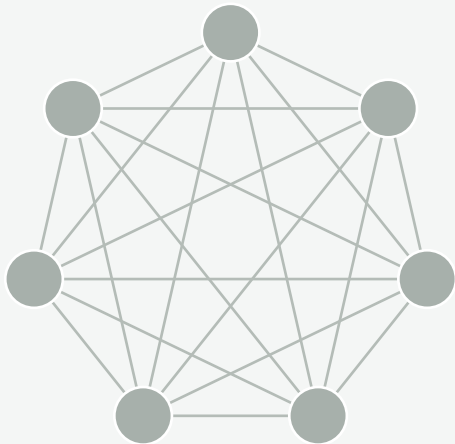
× 6 layers; one-to-many matching;
outputs trees + ground / wood / leaf

INSIDE 3: ONE DECODER LAYER



Why Mamba?

Transformer: Everyone talks to everyone



Like a meeting where every person talks with every other person

2× bigger plot
→ 4× more work

Used by the previous best models: accurate, but slow and memory-hungry on large plots

Mamba: Pass a note down the line

The note is passed along →



Like reading a book once while taking notes

2× bigger plot
→ 2× more work

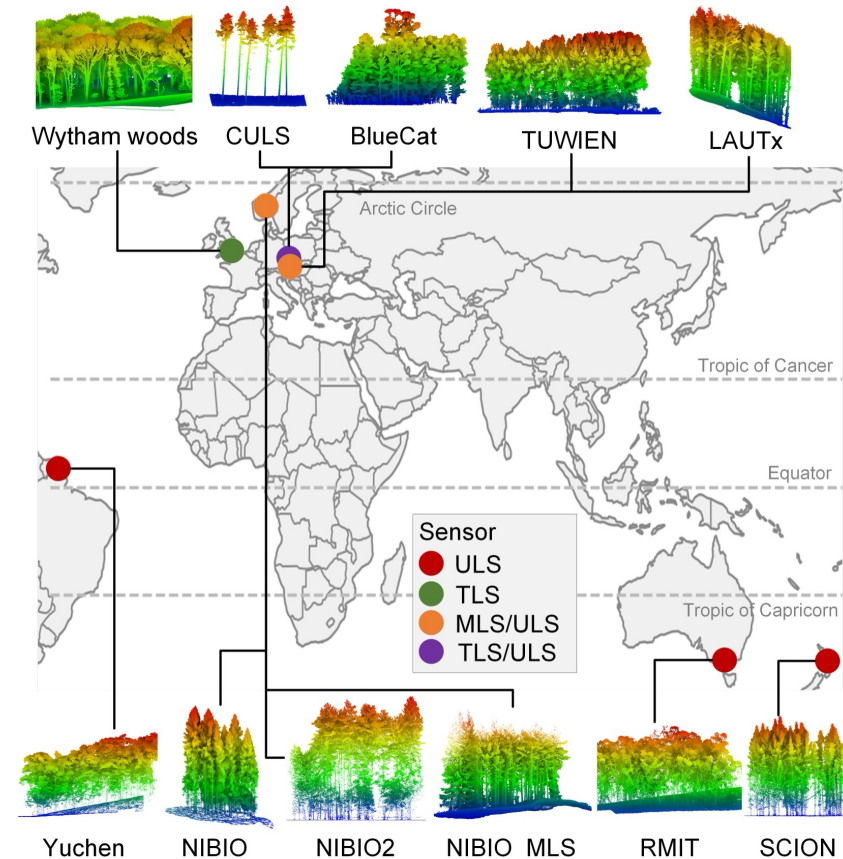
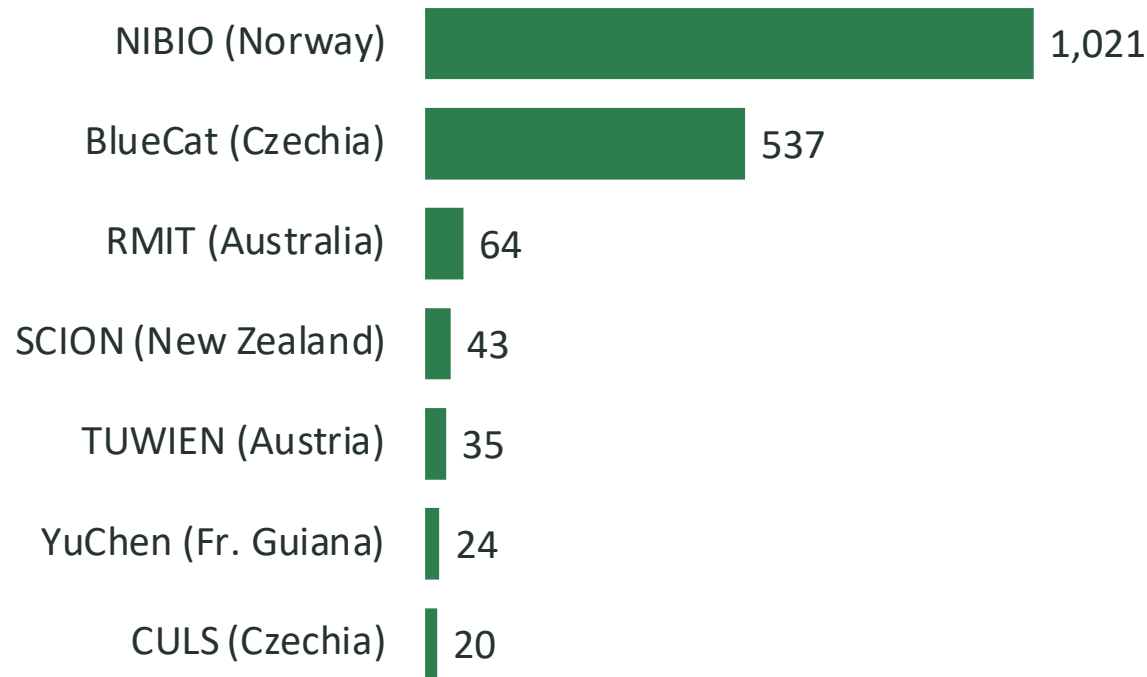
One condition: the reading order must make sense. For trees, we read from the ground up



In plain words: Mamba keeps the model fast even for very large forest plots

How we test it: seven forests, four continents

Test trees per plot region (FOR-instanceV2)



In plain words: Seven very different forests – four continents, from boreal to tropical
Evaluation metrics: Precision, Recall, F1, Coverage, and mIoU

Quantitative results

Method	P	R	F1	Cov.	mIoU
TreeLearn	82.0	36.6	50.6	52.2	—
ForAINet	86.7	65.8	72.8	63.9	82.0
OneFormer3D	82.2	67.3	74.0	63.8	80.8
ForestFormer3D	91.3	74.9	82.3	70.5	82.1
ForestMamba	91.4	76.6	83.4	71.9	82.3

P = Precision; R = Recall; Cov. = Coverage

+1.7

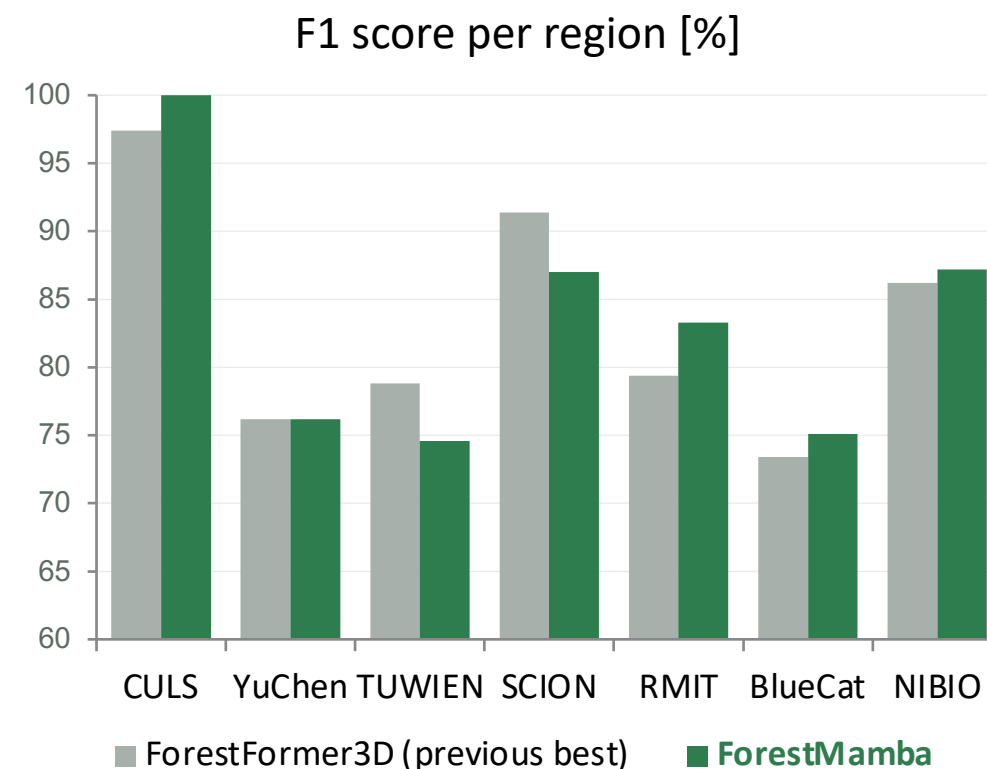
Recall (R): more
real trees found

+0.1

Precision (P): no
extra false trees

+1.1

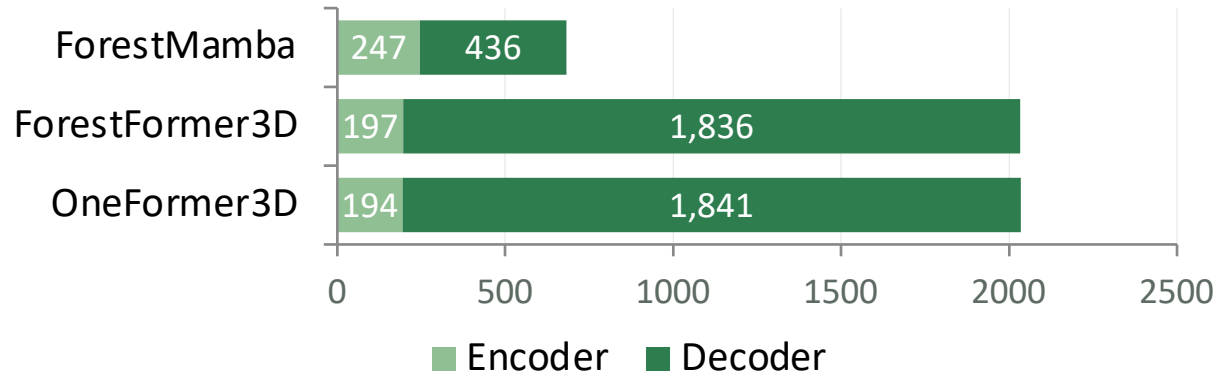
F1 vs. best previous
model



In plain words: ForestMamba finds more of the real trees without adding false ones
Gains are largest in big, dense plots; two small plots (TUWIEN, SCION) are slightly worse

About 3× faster, with less than half the memory

Time per plot [ms] on one RTX A6000 GPU



GPU memory [GB]



2.0 s → 0.7 s

time per plot

28 → 12 GB

GPU memory

4.2×

faster tree outlining

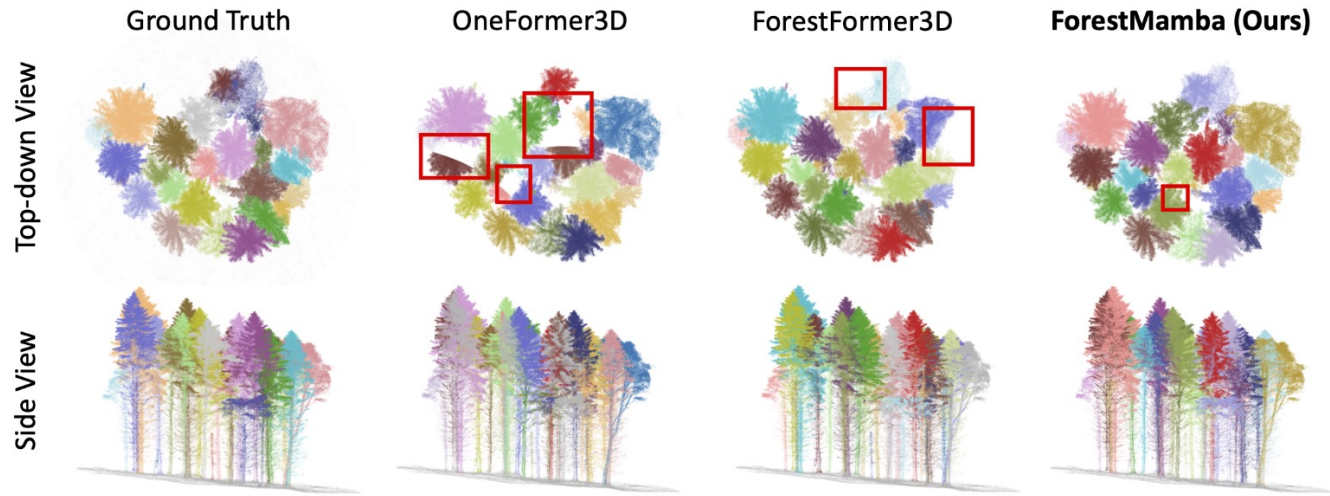
same size

17.3M vs 17.5M
parameters

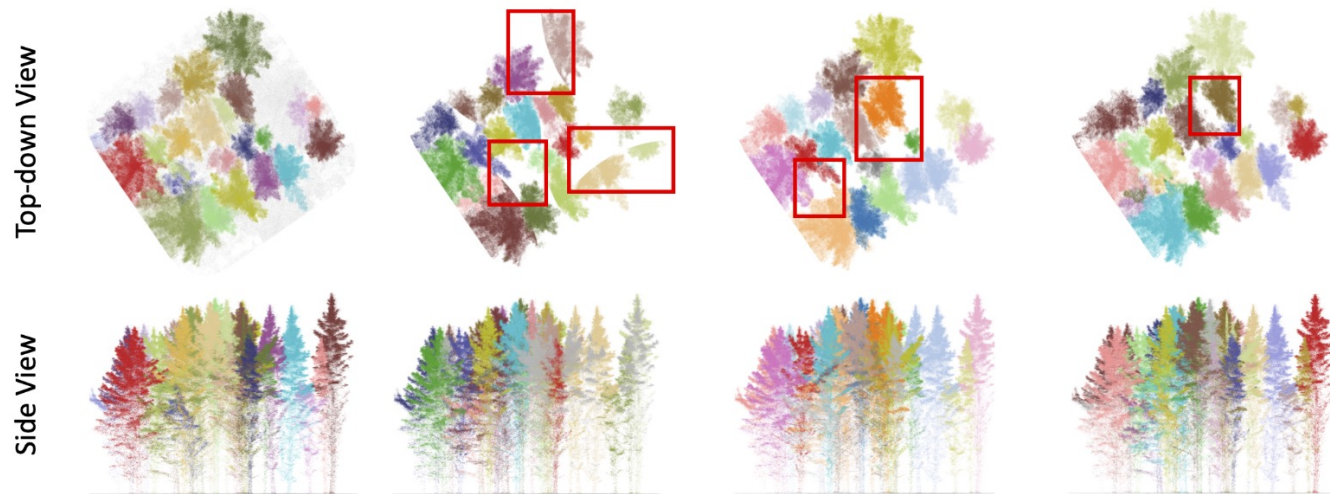


In plain words: On the same GPU, a plot that took about 2 seconds now takes 0.7 seconds, and it needs 12 GB instead of 28 GB of GPU memory

Qualitative results



(a) NIBIO (Norway).



(b) SCION (New Zealand).

What to look for

- **Red boxes:** missing or fragmented trees
- **Previous models:** several errors per scene in dense crowns and partly hidden trees
- **ForestMamba:** one small error per scene, recovers missing trees in top-down view
- **Side view:** stems and crowns are segmented as one coherent instance

Automatic is not enough for annotation



A fixed set of masks

Even at 83 F1, a user cannot pick one tree or repair a wrong one; errors mean manual editing or retraining



Labels are the bottleneck

A hectare has millions of points and hundreds of crowns; per-tree annotation is slow expert work

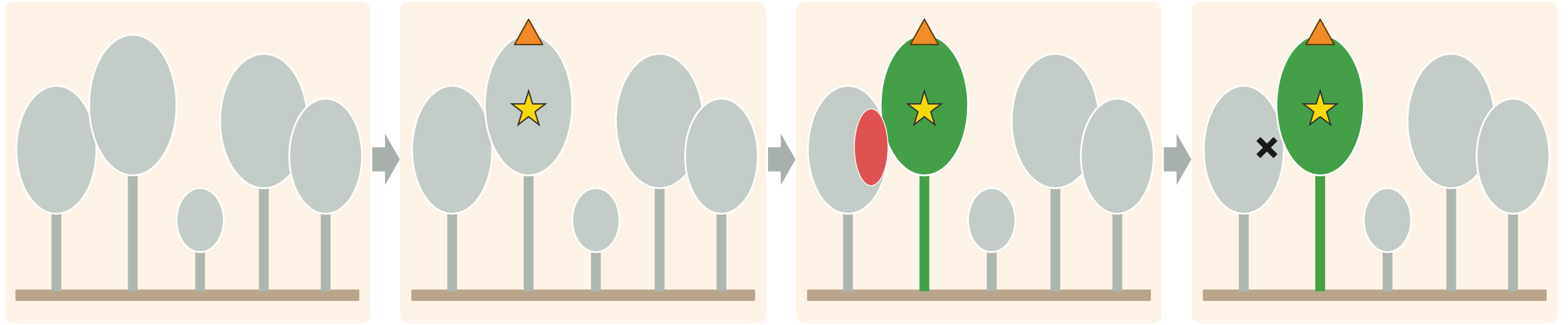


Promptable segmentation

3D versions exist, but for indoor objects, shapes, and parts, not forests

Can a user select, and correct, any tree with a few clicks?

SelectAnyTree: the idea



- 1 Load the plot**
read once,
kept in memory
- 2 Click a tree**
its treetop (▲); optional
- 3 See the tree**
here: a bit too much (red)
- 4 Fix with a click**
× = “not this part”

★ click = “this tree”

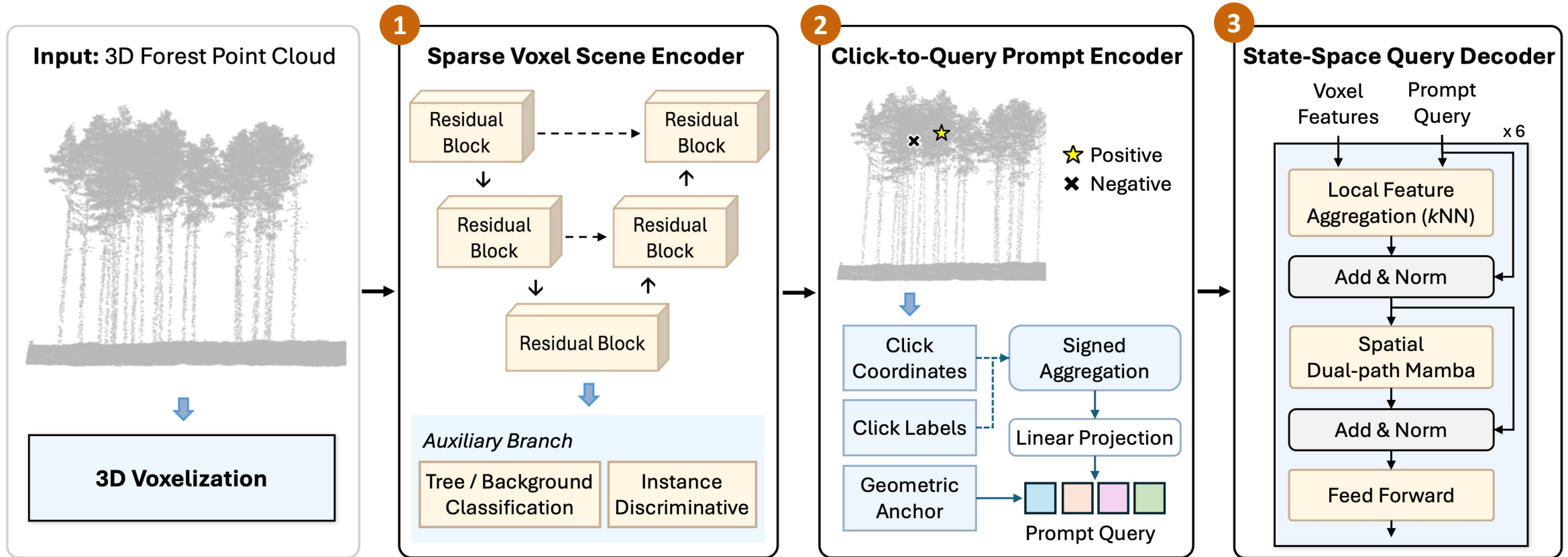
× click = “not this part”

each click gives an answer in a fraction of a second



In plain words: It works like the “magic wand” in photo editing, but for single trees in a 3D forest

SelectAnyTree: from clicks to one tree mask



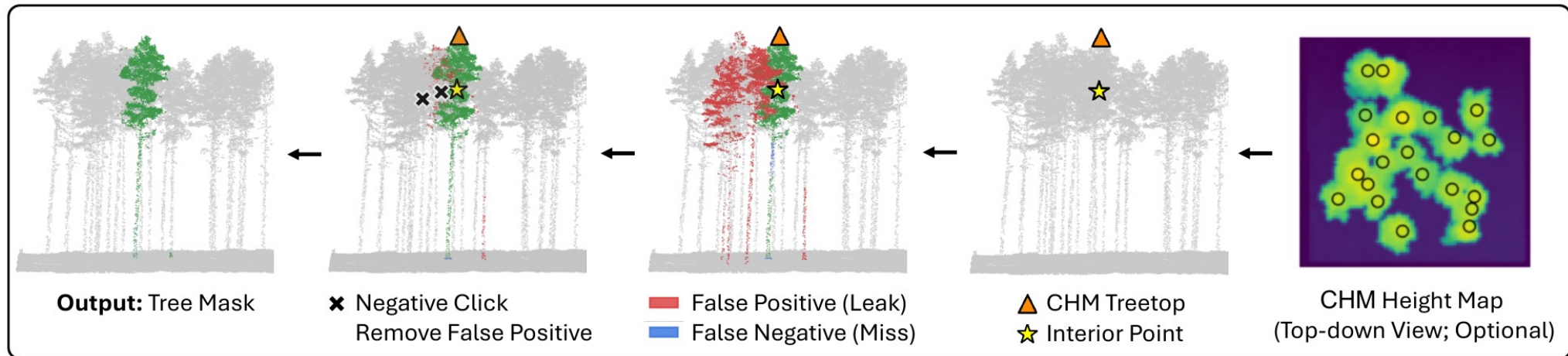
Task Point cloud P , clicks $C \rightarrow$ Tree mask

① Load the plot once $\rightarrow$ ② The click become prompt (**Main of SelectAnyTree**) $\rightarrow$ ③ See the tree

How we test it: a program plays the annotator

- 1 Click the tree centre**
Plus the treetop (Optional)
- 2 Look at the answer**
Compare it with the real tree
- 3 Click the biggest mistake**
★ for a missed part, × for a wrong part; up to 5 clicks

A EXAMPLE (READ FROM RIGHT TO LEFT)



In plain words: The program always clicks exactly where the model is most wrong
Real people are less precise, so these are best-case numbers

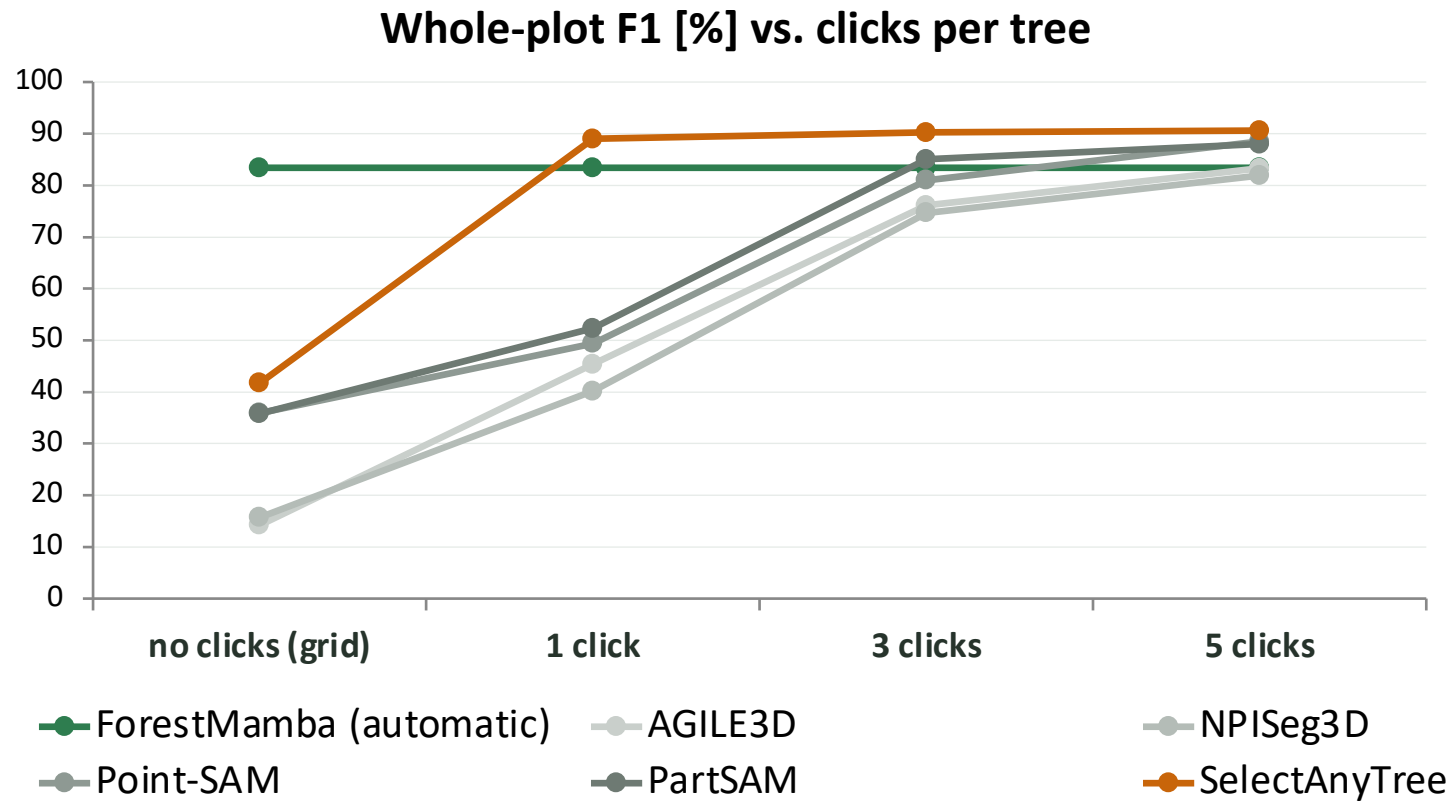
One click → about 80% overlap with the real tree

Method	In-distribution						Cross-dataset Generalization						CHM helps?
	IoU @ Clicks $\uparrow$			NoC @ IoU $\downarrow$			IoU @ Clicks $\uparrow$			NoC @ IoU $\downarrow$			
	1	3	5	0.7	0.8	0.9	1	3	5	0.7	0.8	0.9	
AGILE3D [36]	43.8	64.6	71.9	4.6	5.9	7.8	40.8	62.0	68.7	5.2	6.4	8.3	✗
+ w/ CHM	43.7	59.3	66.7	5.3	6.5	8.3	40.6	56.0	62.5	5.7	7.0	8.8	
NPISeg3D [14]	36.9	61.0	68.5	5.0	6.3	8.1	38.3	59.3	65.1	5.3	6.9	8.8	✓
+ w/ CHM	46.4	63.9	68.3	4.8	6.2	8.2	42.6	61.7	69.3	5.1	6.5	8.5	
Point-SAM [38]	53.4	74.8	80.6	3.4	4.4	6.1	54.5	75.0	81.2	3.3	4.5	6.5	✓
+ w/ CHM	55.2	75.1	81.6	3.4	4.4	6.0	55.7	76.2	81.5	3.2	4.5	6.5	
PartSAM [39]	48.8	71.2	78.3	3.7	4.8	6.6	31.4	53.9	63.9	5.9	7.1	8.5	✓
+ w/ CHM	53.1	74.0	80.8	3.6	4.7	6.8	44.8	66.1	75.5	4.3	5.8	7.8	
SelectAnyTree (Prop.)	77.9	81.3	81.1	3.0	3.9	5.5	76.5	80.3	81.6	2.7	3.7	6.0	✓
+ w/ CHM	79.9	81.9	82.4	2.9	3.8	5.4	77.7	81.5	82.2	2.6	3.4	5.7	



In plain words: With one click, SelectAnyTree outlines about 80% of the tree correctly
Other methods reach about 55%

One click per tree beats the best automatic model



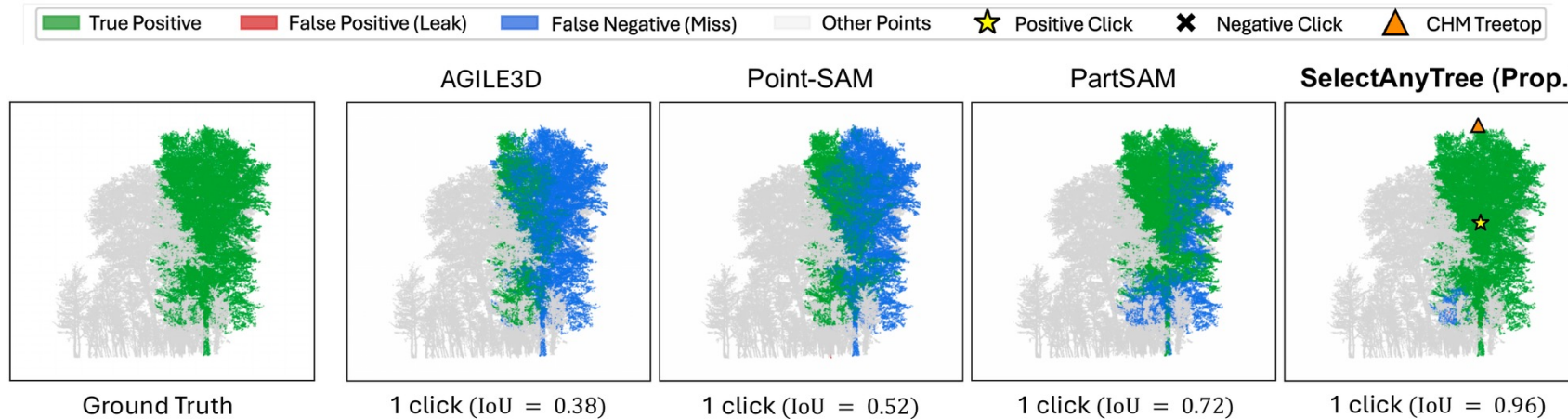
In plain words: Without a person, use the automatic model, **ForestMamba**
If someone can give each tree one click, **SelectAnyTree** gives a better result (89.0 F1)

Smallest and fastest promptable model

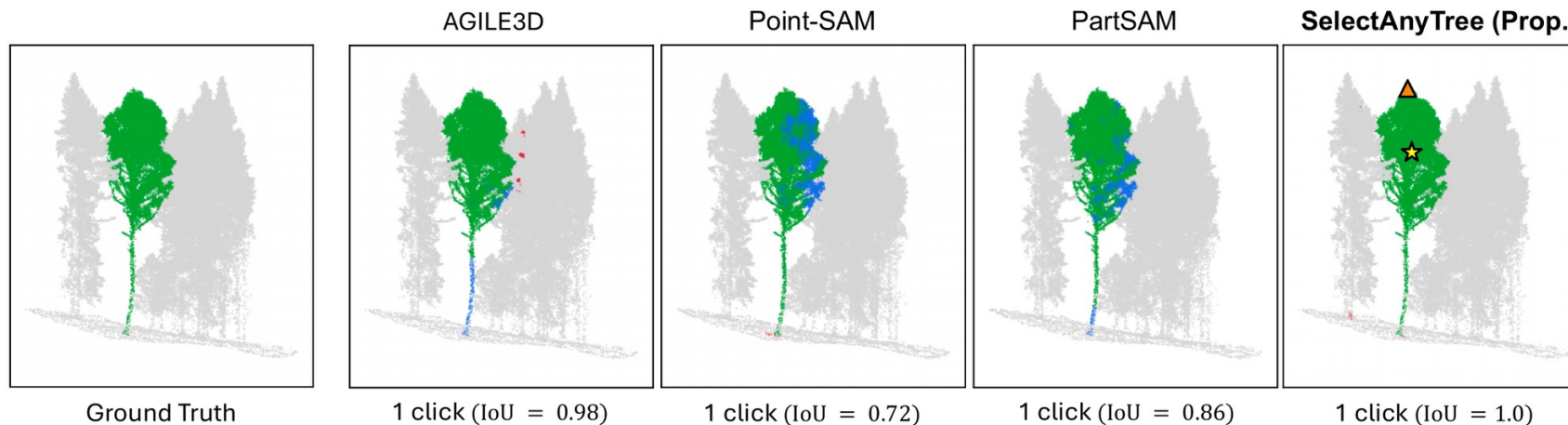
Method	Params [M] ↓	Time [ms] @ Clicks ↓			Peak GPU [GB] ↓
		1	3	5	
AGILE3D [36]	39.3	412 ± 241	1,828 ± 653	3,286 ± 1,043	1.0
NPISeg3D [14]	40.0	249 ± 131	1,307 ± 386	2,353 ± 664	3.1
Point-SAM [38]	311.0	1,612 ± 602	5,062 ± 2,035	8,587 ± 3,472	22.4
PartSAM [39]	225.0	1,549 ± 581	2,558 ± 986	3,562 ± 1,423	2.4
SelectAnyTree (Prop.)	19.4	122 ± 98	298 ± 196	571 ± 184	1.9
+ w/ CHM	19.4	164 ± 159	280 ± 156	549 ± 126	1.9

19.4M**Parameters**16× fewer than
Point-SAM (311M)**571 ms****For 5 clicks**4× faster than NPISeg3D,
15× faster than Point-SAM**1.9 GB****Peak GPU memory**12× less than Point-SAM;
AGILE3D uses 1.0 GB

Qualitative results



(a) BlueCat (Czech Republic) [34]: A dense temperate forest with strongly overlapping crowns.



- **BlueCat (dense):** baselines miss large parts (blue); SelectAnyTree reaches 0.96 IoU
- **NIBIO (conifers):** all methods are closer; only SelectAnyTree gets 1.0
- **Two cues:** the treetop ($\blacktriangle$) anchors the apex, the click ($\star$) grounds the stem

SUMMARY

Two tools that work together

	ForestMamba	SelectAnyTree
In one sentence	Finds every tree and its parts automatically	Let pick and fix any tree with a click
Forest knowledge used	Trees grow upward; treetops mark trees	Treetops mark trees (Optional)
Key result	83.4 F1 · about 3× faster 2.3× less memory	About 80% overlap from 1 click 89.0 F1 with 1 click per tree
Good for	Large automatic forest inventories	Making training labels, fixing errors
Status	BMVC 2026 · on 3Dtrees.earth	Under Review (3DV)



In plain words: ForestMamba for a fast, automatic first pass
SelectAnyTree to check and correct individual trees in a few clicks